



Genômica Computacional

Analise de variantes

Professor: Ricardo A. Vialle

CS31 - Genômica Computacional [27/07-30/07 - 12h00-14h00]

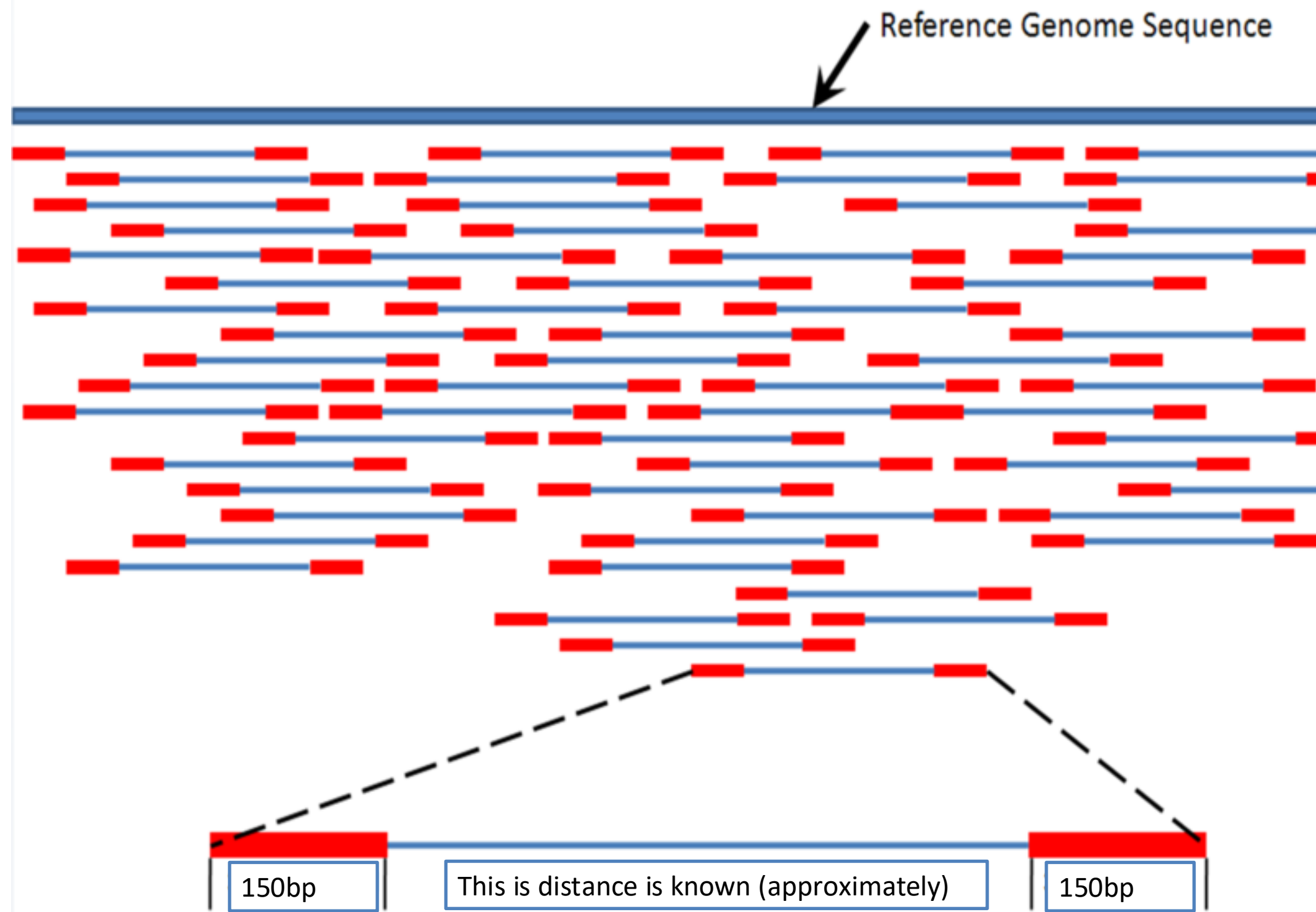
30 de Julho de 2026

Types of genetic variation

- **99% of DNA is shared** between two individuals
- Variation in the remainder explains all our **predisposition** differences
- **Remaining** phenotypic variation: environmental/stochastic differences

Name	Example	Frequency in one genome
Single nucleotide polymorphisms (SNPs)	GAGGAGAACG[C/G]AACTCCGCCG	1 per 1,000 bp
Insertions/deletions (indels)	CACTATTC[C/CTATGG]TGTCTAA	1 per 10,000 bp
Short tandem repeats (STRs)	ACGGCA GTCGTCGTCGTC ACCGTAT	1 per 10,000 bp
Structural variants (SVs) / Copy Number Variants (CNVs)	Large (median 5,000 bp) deletions, duplications, inversions	1 per 1,000,000 bp

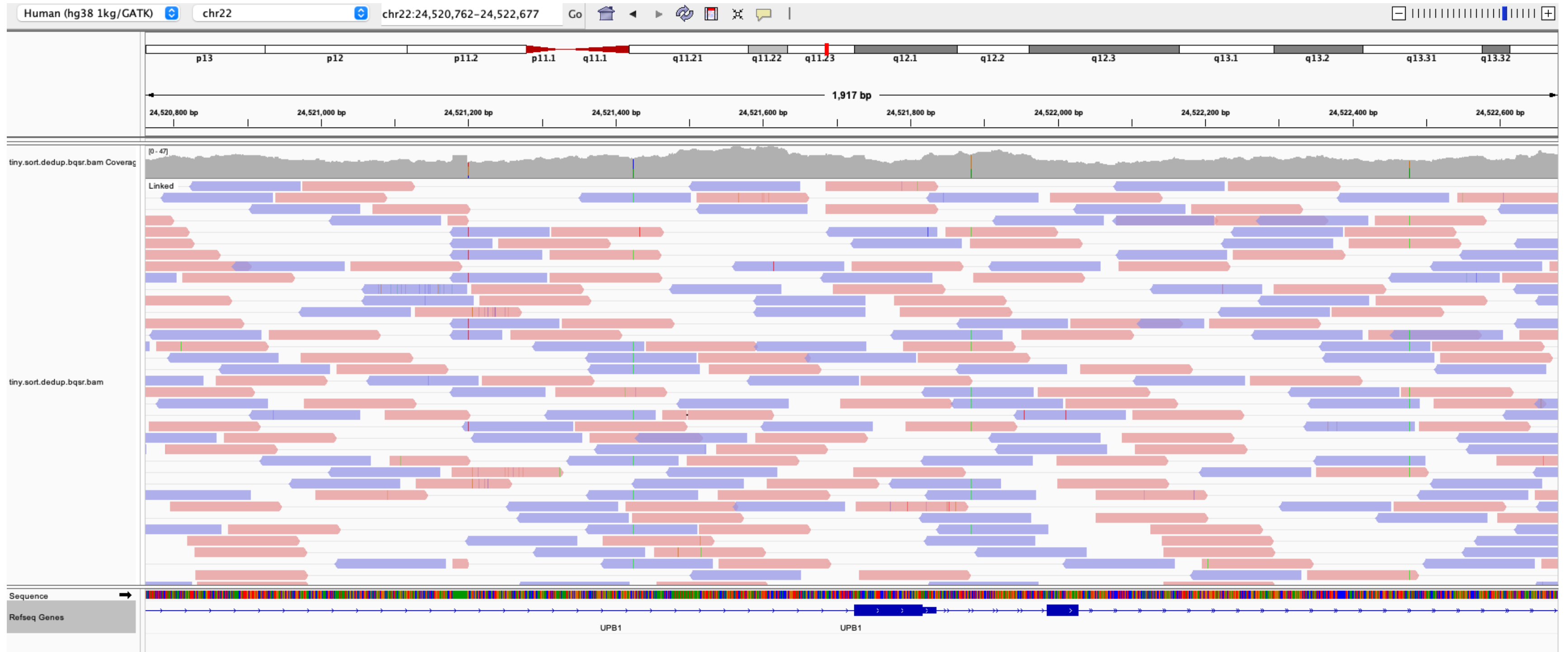
Read mapping



Read mapping

[illegible]

IGV browser



VCF and BCF file format

VCF

```
##fileformat=VCFv4.2
##contig=<ID=2,length=51304566>
##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
```

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1	SAMPLE2	SAMPLE3	SAMPLE4	SAMPLE5	SAMPLE6	SAMPLE7
2	81170	.	C	T	.	.	AC=9;AN=7424	GT:DP:GQ	0/0:4:12	0/0:3:9	0/1:1:3	0/1:9:24	1/0:4:12	0/0:5:15	0/0:4:12
2	81171	.	G	A	.	.	AC=6;AN=7446	GT:DP:GQ	0/1:4:12	0/0:3:9	0/0:1:3	0/0:9:24	0/1:4:12	0/1:5:15	0/0:4:12
2	81182	.	A	G	.	.	AC=5;AN=7506	GT:DP:GQ	0/0:5:15	0/0:4:12	0/0:5:15	0/0:9:24	0/0:4:12	0/0:4:12	0/0:4:12
2	81204	.	T	G	.	.	AC=2;AN=7542	GT:DP:GQ	1/0:5:15	0/0:9:27	0/0:10:30	0/0:15:39	0/0:9:27	1/0:13:39	0/1:14:42

BCF

2	81170	.	C	T	.	.	AC=9;AN=7424	GT:0/0:0/0:0/1:0/1:1/0:0/0:0/0	DP:4:3:1:9:4:5:4	GQ:12:9:3:24:12:15:12
2	81171	.	G	A	.	.	AC=6;AN=7446	GT:0/1:0/0:0/0:0/0:0/1:0/1:0/0	DP:4:3:1:9:4:5:4	GQ:12:9:3:24:12:15:12
2	81182	.	A	G	.	.	AC=5;AN=7506	GT:0/0:0/0:0/0:0/0:0/0:0/0:0/0	DP:5:4:5:9:4:4:4	GQ:15:12:15:24:12:12:12
2	81204	.	T	G	.	.	AC=2;AN=7542	GT:1/0:0/0:0/0:0/0:0/0:1/0:0/1	DP:5:9:10:15:9:13:14	GQ:15:27:30:39:27:39:42

SAM and BAM file format

@HD VN:1.5 SO:coordinate @SQ SN:ref LN:45											Header section
r001	99	ref	7	30	8M2I4M1D3M	=	37	39	TTAGATAAAGGATACTG	*	Alignment section
r002	0	ref	9	30	3S6M1P1I4M	*	0	0	AAAAGATAAGGATA	*	
r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	* SA:Z:ref,29,-,6H5M,17,0;	
r004	0	ref	16	30	6M14N5M	*	0	0	ATAGCTTCAGC	*	
r003	2064	ref	29	17	6H5M	*	0	0	TAGGC	* SA:Z:ref,9,+,5S6M,30,1;	
r001	147	ref	37	30	9M	=	7	-39	CAGCGGCAT	* NM:i:1	

QNAME: query template name, aka. read ID	FLAG: indicates alignment information about the read, e.g. paired, aligned, etc.	RNAME: reference sequence name, e.g. chromosome/transcript id	POS: 1-based position	MAPQ: mapping quality	CIGAR: summary of alignment, e.g. insertion, deletion	RNEXT: reference sequence name of the primary alignment of the NEXT read. For paired-end sequencing, NEXT read is the paired read, corresponding to the RNAME column.	PNEXT: Position of the primary alignment of the NEXT read in the template. Set as 0 when the information is unavailable. It corresponds to POS column.	TLEN: the number of bases covered by the reads from the same fragment. Plus/minus means the current read is the leftmost/rightmost read. E.g. compare first and last lines.	SEQ: read sequence	QUAL: read quality; * meaning such information is not available	Optional fields in the format of TAG:TYPE:VALUE
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SAM and BAM file format (FLAG)

Bitwise Flags

Integer	Binary	Description (Paired Read Interpretation)
1	0000000000001	template having multiple templates in sequencing (read is paired)
2	0000000000010	each segment properly aligned according to the aligner (read mapped in proper pair)
4	0000000000100	segment unmapped (read1 unmapped)
8	000000001000	next segment in the template unmapped (read2 unmapped)
16	000000010000	SEQ being reverse complemented (read1 reverse complemented)
32	000000100000	SEQ of the next segment in the template being reverse complemented (read2 reverse complemented)
64	000001000000	the first segment in the template (is read1)
128	000010000000	the last segment in the template (is read2)
256	000100000000	not primary alignment
512	001000000000	alignment fails quality checks
1024	010000000000	PCR or optical duplicate
2048	100000000000	supplementary alignment (e.g. aligner specific, could be a portion of a split read or a tied region)

Use *samtools flagstat* to summarize these metrics

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r002	0	ref	9	30	3S6M1P1I4M	*	0	0	AAAAGATAAGGATA	*	
r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	* SA:Z:ref,29,-,6H5M,17,0;	
r004	0	ref	16	30	6M14N5M	*	0	0	ATAGCTTCAGC	*	
r003	2064	ref	29	17	6H5M	*	0	0	TAGGC	* SA:Z:ref,9,+,5S6M,30,1;	
r001	147	ref	37	30	9M	=	7	-39	CAGCGGCAT	* NM:i:1	

Optional fields in the format of TAG:TYPE:VALUE

QUAL: read quality; * meaning such information is not available

SEQ: read sequence

TLEN: the number of bases covered by the reads from the same fragment. Plus/minus means the current read is the leftmost/rightmost read. E.g. compare first and last lines.

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CIGAR: summary of alignment, e.g. insertion, deletion

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POS: 1-based position

RNAME: reference sequence name, e.g. chromosome/transcript id

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QNAME: query template name, aka. read ID

CIGAR

For example:

RefPos:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Reference:	C	C	A	T	A	C	T	G	A	A	C	T	G	A	C	T	A	A	C
Read:	ACTAGAATGGCT																		

Aligning these two:

RefPos:	1	2	3	4	5	6	7		8	9	10	11	12	13	14	15	16	17	18	19
Reference:	C	C	A	T	A	C	T		G	A	A	C	T	G	A	C	T	A	A	C
Read:					A	C	T	A	G	A	A		T	G	G	C	T			

With the alignment above, you get:

POS: 5
CIGAR: 3M1I3M1D5M

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r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	* SA:Z:ref,29,-,6H5M,17,0;	
r004	0	ref	16	30	6M14N5M	*	0	0	ATAGCTTCAGC	*	
r003	2064	ref	29	17	6H5M	*	0	0	TAGGC	* SA:Z:ref,9,+,5S6M,30,1;	
r001	147	ref	37	30	9M	=	7	-39	CAGCGGCAT	* NM:i:1	

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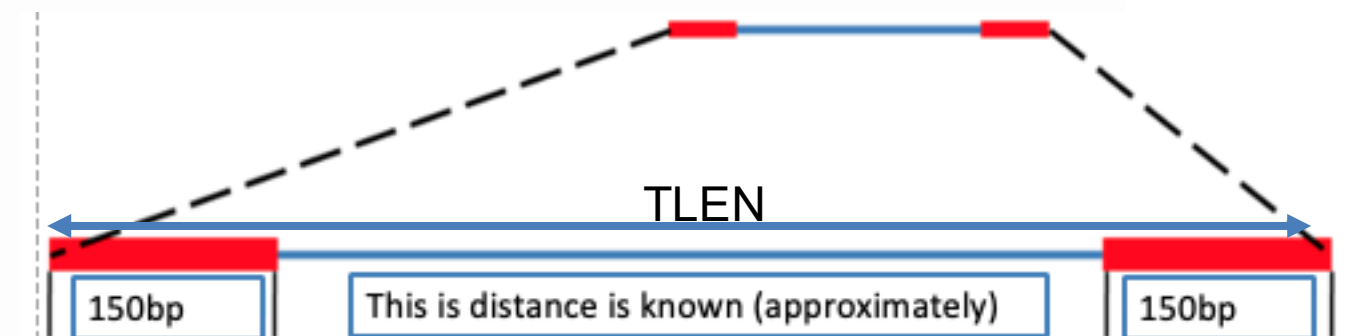
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QNAME: query template name, aka. read ID

SAM and BAM file format

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r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	* SA:Z:ref,29,-,6H5M,17,0;	
r004	0	ref	16	30	6M14N5M	*	0	0	ATAGCTTCAGC	*	
r003	2064	ref	29	17	6H5M	*	0	0	TAGGC	* SA:Z:ref,9,+,5S6M,30,1;	
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</



SAM and BAM file format

QHD VN:1.5 S0:coordinate											Header section
@SQ SN:ref LN:45											
r001	99	ref	7	30	8M2I4M1D3M	=	37	39	TTAGATAAAGGATACTG	*	Alignment section
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r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	* SA:Z:ref,29,-,6H5M,17,0;	
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r003	2064	ref	29	17	6H5M	*	0	0	TAGGC	* SA:Z:ref,9,+,5S6M,30,1;	
r001	147	ref	37	30	9M	=	7	-39	CAGCGGCAT	* NM:1:1	

SAM and BAM file format

Tag ↕	Type ↕	Description [hide] ↕
AM	i	The smallest template-independent mapping quality in the template
AS	i	Alignment score generated by aligner
BC	Z	Barcode sequence identifying the sample
BQ	Z	Offset to base alignment quality (BAQ)
BZ	Z	Phred quality of the unique molecular barcode bases in the OX tag
CB	Z	Cell identifier
CC	Z	Reference name of the next hit
CG	B,I	BAM only: CIGAR in BAM's binary encoding if (and only if) it consists of >65535 operators
CM	i	Edit distance between the color sequence and the color reference (see also NM)
CO	Z	Free-text comments
CP	i	Leftmost coordinate of the next hit
CQ	Z	Color read base qualities

•
•
• SA:Z:ref,29,-,6H5M,17,0;
•
• SA:Z:ref,9,+,5S6M,30,1;
• NM:i:1

Header
section

Alignment
section

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covered by the reads from the same fragment. Plus/minus
leftmost/rightmost read. E.g. compare first and last lines
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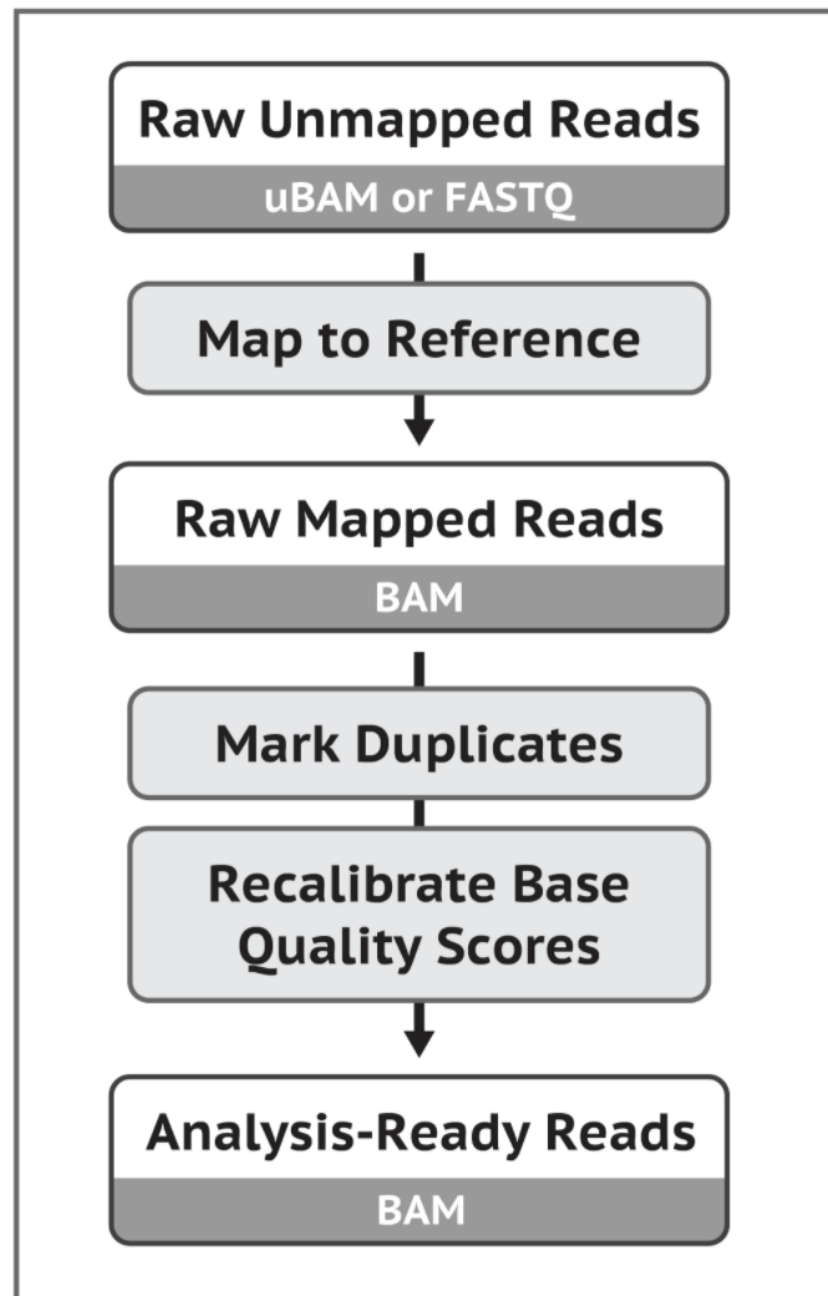
RNAME: reference sequence name, e.g. chromosome/transcript id

FLAG: indicates alignment information about the read, e.g. paired, aligned, etc.

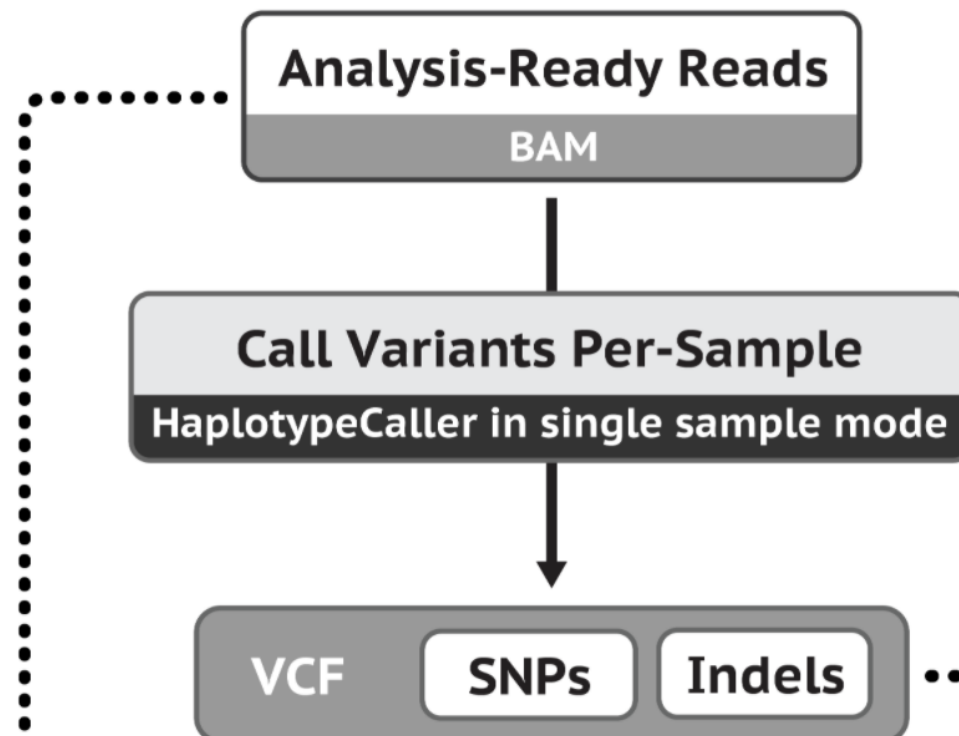
QNAME: query template name, aka. read ID



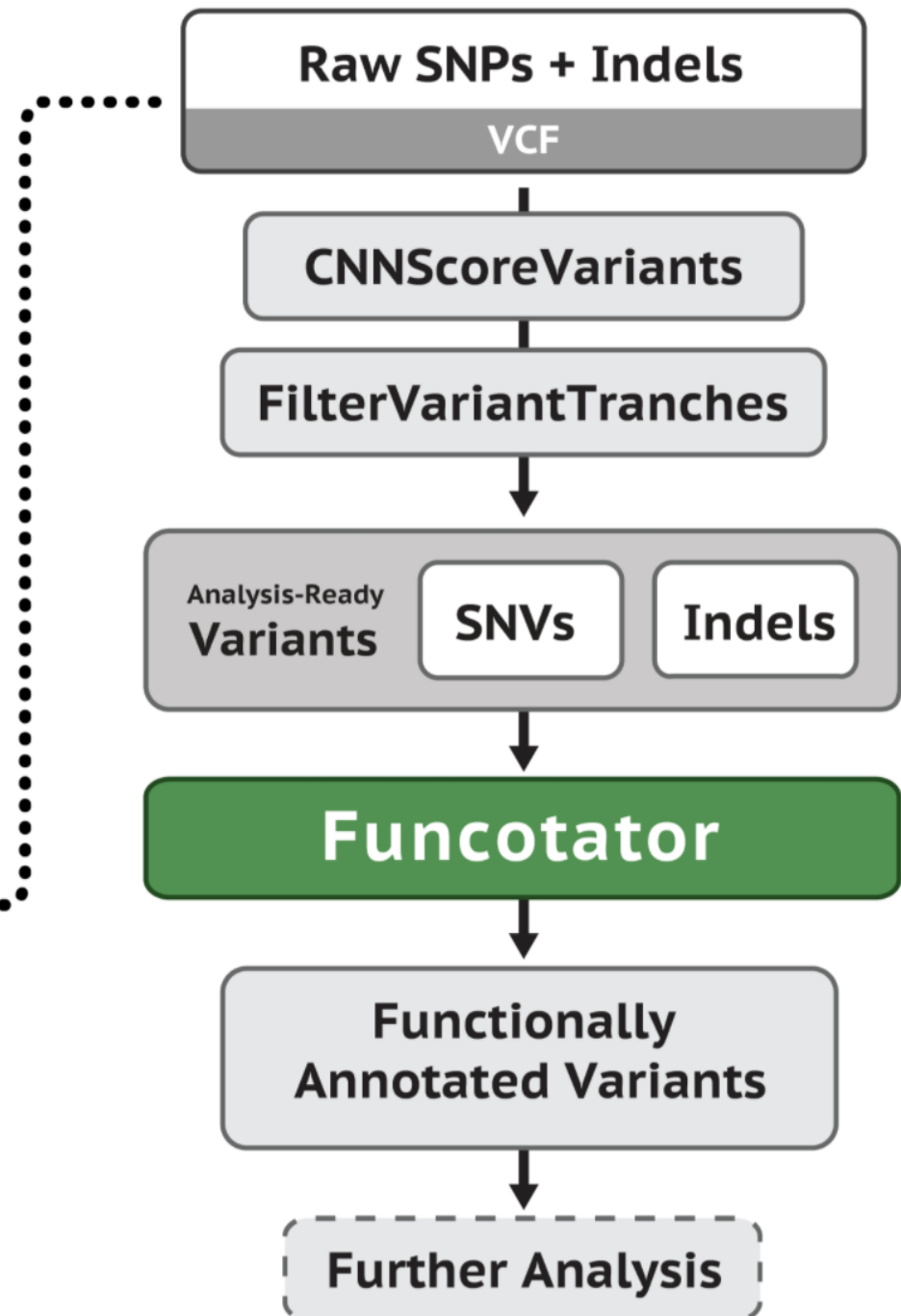
PRE-PROCESSING



VARIANT DISCOVERY



CALLSET REFINEMENT



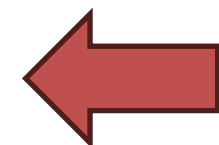
https://github.com/RushAlz/IAMSPE-CS31-Genomica_Computacional

CS31 - Genômica Computacional

Esse repositório contém materiais de aula para a disciplina de Genômica Computacional da IAMSPE.

Para facilitar execução dos tutoriais, utilizaremos o [Google Cloud Shell](#).

Aula	Data	Tema	Slides	Tutoriais
1	2026-07-27	Introdução e processamento de dados de sequenciamento	Slides	Fastq Quality-Control (QC) tutorial
2	2026-07-28	Montagem de genomas	Slides	Genome Assembly tutorial
3	2026-07-29	Anotacao de genomas	Slides	Genome Annotation tutorial
4	2026-07-30	Mapeando variantes	Slides	Variant mapping tutorial



https://github.com/RushAlz/IAMSPE-CS31-Genomica_Computacional

Open in Cloud Shell

You are about to clone the repo:

! https://github.com/RushAlz/IAMSPE-CS31-Genomica_Computacional.git

This repo is not officially maintained by Google and is considered untrusted by default.

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